Harish Manoharan

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Software engineer at Purgo AI working on agents, retrieval and LLM evaluation. M.S. Computer Science, UT Dallas, 2026 (GPA 3.92/4.00); dual degree in Data Science and Biological Sciences, IIT Madras.

EXPERIENCE

Purgo AI San Francisco Bay Area (Los Altos, CA)

AI startup building an agent that writes and validates Databricks and dbt code

Software Engineer Jun 2026 – Present

Software Engineer Intern Jun 2025 – May 2026

- Cut the tables the agent failed to retrieve by 70% on a benchmark built from real tickets by building dbt lineage expansion (sqlglot SQL parsing, ref() and source() resolution); chose the design by ablating four variants and a repo-search alternative.
- Built Purgo's IQ/OQ validation engine for regulated (GxP) customers in NestJS/TypeScript on GCP Cloud Run: it qualifies a customer's Databricks platform on Azure or AWS against a git-versioned library of 59 tests, produces signed evidence reports, and turns uploaded requirement documents into new executable tests through an LLM pipeline.
- Built the LLM validation product (Next.js, NestJS, PostgreSQL) that qualifies Azure OpenAI and Databricks-served models for regulated use, testing hallucination on missing fields, extraction accuracy, demographic bias, determinism, latency under concurrency and traceability, plus the deployment's access, network and audit setup.
- Extended the agent's self-review loop: AST lint rules for generated PySpark/SQL, each derived from a real Databricks runtime failure; dbt source checks that catch a recurring parse failure at review time; and a recovery attempt with best-candidate selection when every candidate fails.
- Built semantic search over Polaris/Iceberg catalogs on Pinecone (incremental re-indexing, keyword fallback) and the benchmark to judge it: about a quarter more of the right tables than the existing search, at about 20x lower latency.
- Reworked attachment chunking so records stay whole and the model is told which parts it didn't get, and moved two stages to a smaller reasoning model after benchmark runs showed no recall loss (7.6x cheaper on those stages).

Digitus Business Solutions Bengaluru, India

Software Engineer Intern May 2024 – Jul 2024

- Raised SQL execution accuracy from 52% to 76% on a 500-query evaluation set by building a text-to-SQL agent with schema discovery and retries driven by execution errors.
- Built a multimodal RAG assistant over internal slide decks and videos (layout-aware chunking, enforced citations).

PwC India Bengaluru, India

Technology Consulting Intern, Data Analytics May 2023 – Jul 2023

- Cut forecast error (MAPE) 15% against the baseline across 50+ SKUs with an XGBoost + SARIMA ensemble, and built aspect-based sentiment analysis of customer reviews.

EDUCATION

The University of Texas at Dallas M.S. Computer Science May 2026

GPA 3.92/4.00 · Dean's Graduate Scholar

Indian Institute of Technology Madras Dual Degree: M.Tech Data Science, B.S. Biological Sciences May 2024

INSPIRE Scholar · Young Research Fellow · Thesis with The Jackson Laboratory on unsupervised behavior discovery in mice

SKILLS

Languages: Python, TypeScript, JavaScript, SQL

Machine learning and LLMs: retrieval-augmented generation (RAG), LLM evaluation, AI agents (LangGraph), embeddings and vector search (Pinecone, pgvector), LLM observability (Langfuse), PyTorch

Data: Databricks, Apache Spark (PySpark), dbt, PostgreSQL, Apache Iceberg

Backend and cloud: FastAPI, NestJS, Next.js, REST APIs, Azure, AWS, GCP, Docker, CI/CD (GitHub Actions)